

VA Ann Arbor Healthcare System
Guidance for Using the Biofire Meningitis Encephalitis PCR Panel
Antimicrobial Stewardship Committee December 2025

The Meningitis/Encephalitis Panel (MEPAN) is a PCR-based test that allows for the simultaneous detection and identification of 14 different pathogens (bacterial, viral, and fungal) in CSF samples. **The MEPAN should always be used as an adjunct to, and not replacement for, conventional CSF testing.**

Recommendations for use:

- Tests may be sent on all patients with suspected acute community-acquired meningitis or encephalitis
- The CSF analysis with reflex to MEPAN can be ordered without ID approval. **ID consult should then be obtained**
- When cryptococcal disease is suspected, direct testing for Cryptococcus antigen from CSF is advised due to higher published sensitivity than the MEPAN
- When HSV meningoencephalitis is suspected, direct testing for HSV in the CSF is advised due to higher published sensitivity than the MEPAN

Pathogens	Empiric Treatment Recommendations*	Notes
<i>Escherichia coli K1</i>	Ceftriaxone 2 g q12h	If cephalosporin allergy: Meropenem 2 g q8h
<i>Haemophilus influenzae</i>	Ceftriaxone 2 g q12h	If cephalosporin allergy: Meropenem 2 g q8h
<i>Listeria monocytogenes</i>	Ampicillin 2 g q4h	If Beta-lactam allergy: Trimethoprim/sulfamethoxazole 5 mg/kg IV q8h OR Meropenem 2 g q8h
<i>Neisseria meningitidis (encapsulated)</i>	Ceftriaxone 2 grams q12h	If cephalosporin allergy: Meropenem 2 g q8h
<i>Streptococcus agalactiae</i>	Penicillin G 24 million units per day	If penicillin allergy: Ceftriaxone 2 g q12h If cephalosporin allergy: Vancomycin 15 mg/kg IV q12h IF susceptible
<i>Streptococcus pneumoniae</i>	Vancomycin 20-25 mg/kg IV loading dose then 15-20 mg/kg IV q12h PLUS Ceftriaxone 2 g q12h PLUS Dexamethasone 10 mg IV q6h	If cephalosporin allergy: Vancomycin dosing (see empiric dosing) OR Linezolid 600 mg IV q12h PLUS Dexamethasone 10 mg IV q6h
Cytomegalovirus	Consult ID	A positive test does not equal active disease
Enterovirus	Supportive care	No effective treatment available
Herpes simplex virus 1	Acyclovir 10 mg/kg q8h	MEPAN is less sensitive than CSF HSV-1 PCR
Herpes simplex virus 2	Acyclovir 10 mg/kg q8h	MEPAN is less sensitive than CSF HSV-2 PCR
Human herpesvirus 6	Consult ID	A positive test does not equal active disease
Human parechovirus	Supportive care	No effective treatment available
Varicella zoster virus	Acyclovir 10 mg/kg q8h	Requires airborne precautions
<i>Cryptococcus neoformans/gattii</i>	Liposomal Amphotericin B 3-5 mg/kg IV PLUS Flucytosine 25 mg/kg PO QID	MEPAN is less sensitive than CSF Cryptococcal antigen

*Adjust dose based on renal function

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References					
	Citation				
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2	Lindström J, Elfving K, Lindh M, Westin J, Studahl M. Assessment of the FilmArray ME panel in 4199 consecutively tested cerebrospinal fluid samples. <i>Clin Microbial Infect.</i> 2022;28(1):79-84. doi:10.1016/j.cmi.2021.05.017				
3	Miller JM, Binnicker MJ, Campbell S, et al. Guide to Utilization of the Microbiology Laboratory for Diagnosis of Infectious Diseases: 2024 Update by the Infectious Diseases Society of America (IDSA) and the American Society for Microbiology (ASM). <i>Clin Infect Dis.</i> Published online March 5, 2024. doi:10.1093/cid/ciae104				
4	Van TT, Kim TH, Butler-Wu SM. Evaluation of the Biofire FilmArray meningitis/encephalitis assay for the detection of <i>Cryptococcus neoformans/gattii</i> . <i>Clin Microbiol Infect.</i> 2020;26(10):1375-1379. doi:10.1016/j.cmi.2020.01.007				
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